UPFRONT

WEATHER: MONSOON JITTERS *PG 10*

VIEWPOINT: DALAI LAMA AND CHINA'S PLAY *PG 12*



MANDAR DEODHAR

FUTURE-READY
A hybrid solar cell lab at IIT Bombay

▶ SCIENCE & TECH

PRIVATE R&D GETS BIG PUSH

The Centre's Rs 1 lakh crore scheme seeks to drive bigger innovation risks by private sector. But it will require breaking free from bureaucratic and *jugaad* culture

By ANILESH S. MAHAJAN

IN BENGALURU'S ELECTRONIC CITY, QNu Labs is refining its quantum key distribution technology—a cybersecurity breakthrough in the making. In Delhi, Vyome Biosciences is using AI to discover new treatments for skin diseases. From tactical drone startups like IdeaForge to space-tech firms such as Skyroot, deep tech ventures across India are pushing boundaries. What's missing is institutional capital and trust for long-gestation research.

India has historically always feared that last gorge which new technology must cross—the "valley of death". It is only the State that can lead the horse across it. But past efforts have been timid. The Technology Development Board, established in the 1990s, was criticised for risk aversion. The Fund of Funds for Start-ups leaned towards low-tech B2C ventures. The Technology Acquisition and Development Fund, under Make in India, barely disbursed 10 per cent of its corpus. A senior industry voice, who has long advised policymakers, says it was not lack of intent, but a failure to grasp the essence of innovation finance. Exciting prototypes often fail for being too commercial for academic grants, too risky for private capital.

Wiser from experience, now comes India's latest initiative to cross that frontier—designed so as to avoid our past behavioural